

When Generative AI Mixes Languages: Multilingual Users' Code-Switching Behavior in Human-LLM Interaction

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Abstract

As large language model (LLM) chatbots are increasingly used by multilingual users, language mixing have become routine practices rather than edge cases. However, prior work has focused primarily on multilingual model performance, with limited attention to how users switch code when interacting with LLMs. Drawing on a mixed-methods study of 84 multilingual users, including surveys, semi-structured interviews, and interaction log analysis, this study examines how multilingual users choose and switch languages conversing with LLMs, and how they interpret the presence of code-switching in AI outputs. Preliminary findings indicate that code-switching is not an incidental behavior but a strategic interactional practice through which users manage task efficiency, linguistic precision, and emotional comfort. At the same time, users report confusion, frustration, and reduced trust when LLMs inconsistently mix languages without clear controls. Based on these findings, we outline design implications for multilingual chatbots that support turn-level language adaptation and enable language-based repair.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction**; • **Empirical studies in HCI**;

Keywords

Generative AI, Human-LLM interaction, Code-switching, Multilingual chatbot

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1 Introduction

Large language models (LLMs) are increasingly embedded in everyday information-seeking, learning, and creative practices, with chat-based interfaces becoming a dominant mode of interaction [17]. As these systems are adopted by users across linguistic and

cultural contexts, multilingual interaction has emerged as a routine yet underexamined aspect of human-LLM communication [4, 10]. While prior work has primarily evaluated LLMs' multilingual performance in terms of accuracy [13, 14], translation quality [6, 16], or cross-lingual generalization [2, 8], considerably less attention has been paid to how users actually use multiple languages when interacting with LLM chatbots.

In everyday LLM use, multilingual users frequently engage in code-switching—the practice of alternating between languages within or across conversational turns—to manage expression, cognition, and task demands [5, 15]. Drawing from sociolinguistics, code-switching has long been understood as a strategic and meaningful communicative practice rather than a sign of linguistic deficiency [9]. However, existing studies on code-switching have largely focused on human-human interaction or multilingual web search [7, 12], but it remains unclear whether these findings extend to interactions between humans and LLMs. As LLMs generate responses that may themselves contain mixed languages, users are required not only to choose how to phrase their inputs, but also to interpret, evaluate, and sometimes correct the model's multilingual outputs [1, 11]. This introduces new interactional challenges that are tightly coupled with interface design rather than language ability alone [3].

Building on a mixed-methods study, this poster investigates *how multilingual users engage in code-switching when conversing with LLM chatbots, how they interpret mixed-language responses from the system, and where breakdowns in understanding and control occur*. Preliminary findings suggest that users dynamically switch languages across and within turns to scaffold expression, preserve domain-specific terminology, repair unsatisfactory outputs, and align responses with intended use contexts. However, participants also report recurring breakdowns, including inconsistent response language inference, degraded translation quality for specialized content, and reduced trust when mixed-language outputs appear unintentional or opaque. By foregrounding users' code-switching behavior in human-LLM interaction, this work reframes language not merely as content but as an interactional resource and an interface concern, and highlights concrete opportunities to design chatbot UIs that better support multilingual communication, control, and user agency.

2 Methods

We employed a mixed-methods design that combined an online survey, semi-structured interviews, and analysis of interaction chatlogs. By triangulating across these data sources, we captured both



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self-reported usage patterns and situated interactional practices, offering a more comprehensive account of multilingual human–LLM interaction.

2.1 Data collection

The online survey was mainly posted in the international students chat groups and other social media platforms. In total, we collected responses from 84 multilingual users who regularly interacted with LLM chatbots. The survey gathered demographic information, language backgrounds, LLM usage frequency, language choice preferences, and self-reported code-switching behaviors. To maximize participants' linguistic diversity, no restrictions were imposed on age, gender, or occupation. The sample consisted of predominantly young adults (mean age = 23.67; median = 23; range = 18–36), with females accounting for 60.7% of participants and males for 39.3%. Participants were primarily from China (79.76%), followed by Malaysia (17.86%), with smaller representations from Mongolia (1.19%) and South Korea (1.19%). In terms of educational background, most respondents held an undergraduate degree (78.57%), while 16.67% had a master's degree; a small proportion reported associate-level (2.38%) or doctoral-level education (2.38%). All participants were proficient in two or more languages. Mandarin Chinese was the dominant first language (94.05%), and English was the most common second language (89.29%), indicating a primary Chinese–English bilingual profile. Malay (35.29%), Japanese (23.53%), Korean (14.71%), and Cantonese (11.76%) emerged as the frequently reported third language.

To complement survey data with in-depth qualitative insights, we conducted semi-structured interviews with 20 participants who volunteered for follow-up sessions. Interviews focused on participants' concrete experiences of code-switching within LLM conversations, including when and why they switched languages, how they interpreted mixed-language responses generated by the system, and what breakdowns they encountered. Interviews lasted around 30 minutes, were audio-recorded with consent, and were transcribed for analysis. In addition, we collected user interaction logs in the form of 67 conversation screenshots provided by interview participants, which naturally captured occurring multilingual exchanges between users and LLMs.

2.2 Data analysis

Quantitative survey data were analyzed using descriptive statistics to characterize overall usage patterns and language-switching frequencies. Qualitative interview transcripts and interaction logs were analyzed using thematic analysis, focusing on language choice rationales, interpretive strategies for mixed-language responses, and moments of interactional breakdown or repair. All study procedures were reviewed and approved by the authors' institutional review board (IRB), and informed consent was obtained from all participants prior to data collection; participants were assured of confidentiality, and all identifiable information was anonymized during transcription and analysis. For the qualitative analysis, the research team followed an iterative coding process: an initial open-coding phase was conducted to inductively identify concepts from the interview transcripts and screenshots, followed by axial coding to group related codes into higher-level categories and themes. The

codebook was refined through repeated team discussions, with discrepancies resolved through consensus to enhance analytic rigor and interpretive reliability.

3 Findings

3.1 Users' code-switching behavior in human-LLM interaction

First, participants reported using LLMs such as DeepSeek (58), ChatGPT (54), Doubao (43), ERNIE (30), Kimi AI (23), Gemini (6), Qwen (3), ChatGLM (3), Claude (2), Character AI (1), Yuanbao (1), Grok (1), and Copilot (1). Among the 84 participants surveyed, 92.9% see themselves as heavy users of LLMs and 96.4% reported switching or mixing languages during LLM conversations, often repeatedly within a single interaction (75%). Language choice was not fixed at the conversation level; instead, users dynamically adjusted languages across turns to manage expression, comprehension, and task requirements. The most influential factor shaping initial language choice was users' familiarity with a language (84.5%), followed by domain-specific conventions (e.g., technical or academic content) (58.3%) and perceptions of the model's language strengths (44%) (see Table 1).

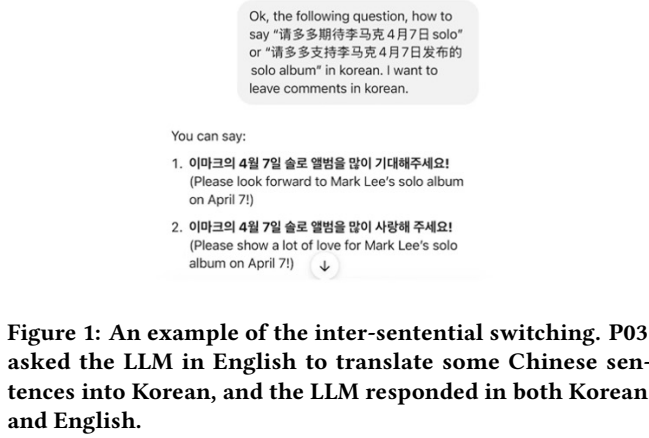
We identified three dominant forms of code-switching in human–LLM interaction: inter-sentential switching, intra-sentential switching, and tag or term insertion. Inter-sentential switching refers to language shifts that occur between complete sentences or clauses, with each unit remaining internally monolingual; this form is often associated with topic shifts, stance marking, or audience alignment (see Figure 1). Intra-sentential switching occurs within a single sentence or clause and involves integrating of lexical items or phrases from multiple languages into a unified syntactic structure, requiring a high degree of bilingual competence and grammatical coordination. This was widespread across everyday queries and was often used to compensate for vocabulary gaps or to preserve domain-specific terminology (see Figure 2). Tag insertion denotes the embedding of fixed discourse markers, pragmatic tags, or formulaic expressions from one language into an otherwise monolingual utterance; these elements are syntactically peripheral and do not alter the grammatical structure of the host sentence. For example, P03 started the conversation with “I’m sorry but” in English and continued the request in Chinese and P17 asked “would you miss me if I don’t talk to you for a while” in Chinese but added “I don’t know” in English at the end of sentence. In these cases, “I’m sorry but” and “I don’t know” are tag insertions.

3.2 User's strategy of switching code in human-LLM interaction

The findings indicate that code-switching is not a marginal or accidental phenomenon in human–LLM interaction, but rather a strategic communicative practice through which users actively manage efficiency, accuracy, and affective comfort. Drawing on interview data, users' code-switching behaviors can be analytically grouped into three interrelated domains: user-driven strategies, system-driven adjustments, and language-structural negotiations. Across these domains, code-switching functions as a form of adaptive interactional work, allowing users to align their linguistic choices

Table 1: Factors influencing initial language choice

	Frequency	Percentage
Native language/language familiarity	71	84.5%
Domain-specific content	49	58.3%
Model's language capability (training data)	37	44%
Cultural relevance	35	41.7%
Personal preference	28	33.3%
Privacy considerations	3	3.6%



with task demands, perceived system capabilities, and the limits of cross-linguistic equivalence.

At the user level, code-switching is primarily motivated by pragmatic considerations such as simplifying interaction and matching the final output language. A majority of participants reported deliberately switching to the target output language—most commonly English—to avoid additional translation steps and to ensure the usability of generated content in academic or professional contexts. As P10 said, “If I know I want English content, I will ask in English.” Relatedly, translation-oriented code-switching was widely

observed, with users treating the LLM as a multilingual translation and post-editing tool by pasting foreign-language texts, links, or images and requesting localized outputs. Differences in individual language proficiency further shaped these practices: when users lacked fluency or lexical confidence in one language, they often employed intra-sentential mixing from a stronger language to scaffold their expression. As P9 explained, “Because I have completed my undergrad education abroad, much of my academic training has been conducted in English. As a result, I am sometimes unsure how to express certain concepts in Chinese.” Habitual language use and emotional expression also played a role, as some users reported switching languages to emphasize key points, convey stance, or create emotional distance when discussing sensitive or personal topics. As P11 elaborated: “I would tell it things that make me feel very low emotionally, such as being criticized by my supervisor at work. These are things I usually wouldn’t share with friends or family, but I would tell ChatGPT. Sometimes I think this is because my native language is Chinese, and responding in Chinese makes it feel like I’m communicating with a real person. So I choose to chat in English.”

System-related factors further influenced users’ code-switching strategies. Participants frequently described switching languages in response to dissatisfaction with initial outputs, particularly when responses were perceived as inaccurate, vague, or misaligned with task intent. In such cases, code-switching operated as a repair mechanism, enabling users to issue more precise instructions or invoke a language in which they believed the model performed better. Notably, several users explicitly articulated assumptions about LLMs’ “native” or dominant languages, most often English, and adjusted their input language accordingly to improve fluency, coherence, or responsiveness. As P7 noted, “Because the chatbot’s native language is English, I communicate with it using a mix of English and Chinese. I do this because I feel that the chatbot’s responses in Chinese are not very good.” This accommodation underscores that users are not passive recipients but active theorists of model competence, dynamically adapting their linguistic practices to optimize interaction quality.

Finally, language-structural factors, particularly cross-linguistic non-equivalence, were a persistent driver of code-switching. Users reported switching languages when encountering semantic gaps, untranslatable concepts, or discipline-specific terminology that lacked stable or widely accepted equivalents. Professional jargon, abbreviations, and culturally embedded expressions were often retained in their original language to preserve precision and efficiency, while surrounding discourse was produced in another language for readability. As P8 remarked: “For example, when dealing with

transcript evaluation, especially for U.S. credential evaluation, it is difficult to translate the term precisely into Chinese. So I would directly input that my transcript requires a Course-by-Course Evaluation, and ask whether I could be granted an extension for submitting the transcript." In these cases, code-switching functioned as a resource for negotiating meaning across linguistic systems rather than as a sign of linguistic deficiency.

3.3 Breakdowns in language inference and output alignment

Despite generally positive evaluations of LLM multilingual capabilities, participants reported recurring breakdowns in interaction. A central source of frustration was the system's inconsistent handling of mixed-language input, particularly when LLMs failed to infer the intended response language and defaulted to a previous conversational state. For example, P13 observed that, "I think the biggest issue is that when I sometimes send it a mix of Chinese, English, or even another minor language, it doesn't always get which language it is supposed to respond in." Limited sensitivity to language-switching cues can require participants to repeatedly specify their preferred response language, thereby increasing interactional friction and the time cost of communication.

Additional breakdowns were reported in translation quality, including overly literal translations, incorrect word order, and imprecise handling of specialized terminology, which reduced trust in outputs for academic or professional use. P15 noted: "*Most of the materials I read are journal articles, which contain specialized terminology. The translations produced by the model often differ from the established Chinese expressions used within our academic community.*" Moreover, text generated by LLMs may at times appear rigid or mechanical, with a discernible 'AI-like' tone and a lack of the natural fluency characteristic of human writing. Beyond deficiencies in linguistic competence, these issues also point to biases in training data and shortcomings in cultural adaptation.

4 Design implications

The findings suggest that multilingual interaction with LLM chatbots should be treated as an interactional design problem rather than a purely linguistic or backend capability. Users actively employ language switching as a resource to simplify communication and controlling outputs, yet current chatbot interfaces provide limited support for making these practices visible and legible. The following implications highlight opportunities for designing chatbot interfaces that better serve multilingual users in human-LLM interactions.

For example, LLM interfaces should treat language choice as dynamic rather than fixed at the conversation level. Systems should continuously infer and adapt to users' language preferences across turns, particularly following explicit or implicit code-switching cues. Also, interfaces should provide lightweight mechanisms, such as persistent response-language indicators or one-click language toggles, that allow users to confirm or adjust output language without repeated re-prompting. In addition, mixed-language input should be interpreted as a meaningful communicative strategy rather than noise. Systems should preserve users' intra-sentential mixing and

term insertions, especially for domain-specific vocabulary, instead of enforcing full translation or normalization.

Furthermore, since users often switch languages to correct or refine system behavior, LLMs should treat language shifts as potential repair signals and adjust interpretation accordingly, rather than resetting conversational assumptions. For professional and academic tasks, systems should selectively retain untranslated technical terms while adapting surrounding text, supporting precision without sacrificing readability. Language choice is not only instrumental but also emotional. Systems should be sensitive to language shifts used for affective distancing, emotional disclosure, or stance marking, and avoid overly rigid or mechanical responses that undermine relational comfort.

5 Discussion and Conclusion

Our findings demonstrate that multilingual users actively theorize about LLM language competence and adapt their linguistic behavior accordingly. Language choice functions as a form of interactional control, allowing users to steer the system and negotiate meaning rather than passively consume AI-generated content. In this sense, code-switching functions as a form of interactional work through which users align their intentions with system affordances. Meanwhile, participants' interpretations of mixed-language AI responses play a critical role in developing code-switching strategy and shaping trust with LLMs. Importantly, users do not treat mixed-language outputs as inherently problematic. Instead, frustration arises when language mixing appears uncontrolled or misaligned with user intent, such as when the system defaults to an unintended response language or inconsistently translates specialized terminology. These breakdowns highlight a gap between users' sophisticated language strategies and interfaces that treat language as a static session-level setting. By situating code-switching within human-LLM interaction, this work contributes a user-centered perspective on multilingual AI interaction, underscoring the importance of treating language as an interface design concern and advancing HCI scholarship on conversational interfaces, multilingual interaction, and user agency.

This study has limitations that also point to avenues for future research. First, the participant sample is relatively limited in size and linguistic diversity, consisting primarily of highly proficient bilingual or multilingual users with Chinese and English as their dominant languages, which may shape both language-switching practices and expectations of LLM behavior; future studies should therefore include users with broader and more heterogeneous multilingual repertoires. In addition, the analysis relies largely on self-reported accounts and retrospective interaction logs, which capture users' interpretations but may not fully reflect real-time cognitive and interactional processes; controlled experiments, in-situ logging, and techniques such as eye-tracking could offer more fine-grained evidence of how users detect and respond to mixed-language outputs as interactions unfold.

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